2-24-26 (WEEK 1)

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GRADING

HW – 30%

Midterm – 30% (Apr 16)

Final – 40% (Jun 11)

COMBINATORIAL ANALYSIS (COUNTING)

BASIC PROBABILITY MODEL

① Sample Space (S): set of all possible outcomes

② Event ($E \subseteq S$): set of outcomes we are interested in

③ Probability: $\mathbb{P}(E) \in [0, 1]$ (uniform model, i.e. w/o bias $\Rightarrow$ $\mathbb{P}(E) = \frac{|E|}{|S|}$)

EXAMPLES

① Rolling a fair die: What's the probability of getting a six?

$S = \{1, 2, 3, 4, 5, 6\}$, $E = \{6\}$

$\Rightarrow \mathbb{P}(E) = \frac{|E|}{|S|} = \frac{1}{6}$

② Rolling a fair die: What's the probability of rolling a prime?

$S = \{1, 2, 3, 4, 5, 6\}$, $E = \{2, 3, 5\}$

$\Rightarrow \mathbb{P}(E) = \frac{|E|}{|S|} = \frac{3}{6} = \frac{1}{2}$

③ Standard Deck of 52 Cards: Draw a random card, what's the probability of getting an ace?

uniformly random unless otherwise specified

$S = \{A\diamondsuit, 2\diamondsuit, 3\diamondsuit, \ldots, A\clubsuit, 2\clubsuit, \ldots, A\heartsuit, 2\heartsuit, \ldots, A\spadesuit, \ldots\}$

$E = \{A\diamondsuit, A\clubsuit, A\spadesuit, A\heartsuit\}$

$\Rightarrow \mathbb{P}(E) = \frac{|E|}{|S|} = \frac{4}{52} = \frac{1}{13}$

④ Rolling two fair die: What's the probability of the sum being 7?

☆ We cannot just do $\{2, 3, 4, \ldots, 10, 11, 12\}$ = sum of dice as our S because they aren't uniformly likely events

$S = \{(i, j) \mid i = 1, \ldots, 6,\ j = 1, \ldots, 6\}$ ← ordered set of rolls

$E = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$

Alternate notation: $i \in [6]$, $j \in [6]$

$\Rightarrow \mathbb{P}(E) = \frac{|E|}{|S|} = \frac{6}{36} = \frac{1}{6}$

COUNTING RULES

SUM RULE

$S_i \cap S_j = \emptyset \ \forall 1 \le i, j \le n,\ i \ne j$

If a set S is a disjoint union $S_1 \sqcup \ldots \sqcup S_n$, then $|S| = \sum_{i=1}^{n} |S_i|$

EXAMPLE

Roll two fair die. What's the probability of at least one odd die?

$E = \{\text{at least one odd die}\} = E_1 \cup E_2$, where $E_1 = \{\text{first die is odd}\}$, $E_2 = \{\text{second die is odd}\}$

However, $E_1 \cap E_2 = \{\text{both die are odd}\} \ne \emptyset$, so we cannot have $|E| = |E_1| + |E_2|$

Correct approach:

$E_1' = \{\text{first die is odd}\}$, $E_2' = \{\text{first die is even and second die is odd}\}$

Then, $E = E_1' \sqcup E_2' \Rightarrow |E| = |E_1'| + |E_2'|$

Notice, $S = \{(x, y) \mid x, y \in [6]\}$, $E_1' \subseteq S = \{(x, y) \mid x \in \{1, 3, 5\},\ y \in [6]\}$, $E_2' \subseteq S = \{(x, y) \mid x \in \{2, 4, 6\},\ y \in \{1, 3, 5\}\}$

$\therefore |E| = |E_1'| + |E_2'| = 18 + 9 = 27$, $\mathbb{P}(E) = \frac{|E|}{|S|} = \frac{27}{36} = \frac{3}{4}$

PRODUCT RULE

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If a set S is the Cartesian product of sets $S_1, S_2, \ldots, S_n$, i.e. $S = S_1 \times S_2 \times \ldots \times S_n = \{(a_1, \ldots, a_n) \mid \forall i,\ a_i \in S_i\}$, then $|S| = \prod_{i=1}^{n} |S_i|$

(Informally, if a big choice can be broken down into a sequence of smaller choices, then the total # options = product of # options of smaller choices)

EXAMPLE

Roll two fair dice. What's the probability that the sum is odd?

$S = \{(x,y) \mid x, y \in [6]\} = S_1 \times S_2$, $S_1 = S_2 = [6] \Rightarrow |S| = |S_1|\,|S_2| = 6(6) = 36$

$E = \{\text{sum is odd}\} = \{\text{exactly one of the two die is odd}\} = E_1 \cup E_2$, where:

↳ $E_1 = \{1^{st} \text{ odd}, 2^{nd} \text{ even}\} = \{(x,y) \mid x \in \{1,3,5\}, y \in \{2,4,6\}\}$

↳ $E_2 = \{1^{st} \text{ even}, 2^{nd} \text{ odd}\} = \{(x,y) \mid x \in \{2,4,6\}, y \in \{1,3,5\}\}$

Then,

- $E_1 = \{1,3,5\} \times \{2,4,6\} \Rightarrow |E_1| = |\{1,3,5\}| \times |\{2,4,6\}| = 3(3) = 9$
- $E_2 = \{2,4,6\} \times \{1,3,5\} \Rightarrow |E_2| = 3(3) = 9$

By the sum rule, $|E| = |E_1| + |E_2| = 9 + 9 = 18$

$\therefore \mathbb{P}(E) = \frac{|E|}{|S|} = \frac{18}{36} = \frac{1}{2}$

ADVANCED PRODUCT RULE

If we are making a series of n choices, and for the ith choice, we always have k_i choices available, then the total # choices $= \prod_{i=1}^{n} k_i$

EXAMPLE

Roll two fair die, what's the probability the sum is odd?

$S = [6] \times [6] \Rightarrow |S| = 36$, $E = \{(x,y) \mid x+y \text{ is odd}\}$

Notice:

- Which die is odd? 1st or 2nd
- If 1st die is odd, then 2nd has 3 options
- If 2nd die is odd, then 1st has 3 options

$\therefore |E| = 3(3) = 9$

PERMUTATIONS

Claim: There are $n!$ ways to order n distinct objects

Proof: Use the Advanced Product Rule

↳ Which element comes first? #options $= n$

↳ Which element comes second? #options $= n-1$

⋮

↳ Which element comes i^{th}? #options $= n-i+1$ $(1 \leq i \leq n)$

$\therefore$ By Advanced Product Rule, total # of orderings $= \prod_{i=1}^{n} n-i+1 = n!$ □

COMBINATIONS

How many subsets of size r we choose from an n-element set?

(i.e. How many ways are there to choose r elements out of n without order?)

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DEFINITION

The binomial coefficient $\binom{n}{r}$ ("n choose r") counts the # r-element subsets of an n-element set (Note: We are defining $\binom{n}{r}$ as the sol to this problem, not as $\frac{n!}{r!(n-r)!}$)

CLAIM

$\forall\, 0 \le r \le n,\ \binom{n}{r} = \frac{n!}{r!(n-r)!}$

PROOF

We count the number of ways of counting all n items in two different ways (double counting)

① Direct Permutation of n items $\rightarrow n!$

② Step 1: Choose the elements in the first r elements (unordered)

Step 2: Order the r elements

Step 3: Order the remaining n-r elements

By the Advanced Product Rule, # permutations $= \binom{n}{r} r!\,(n-r)!$

$\therefore\ n! = \binom{n}{r} r!\,(n-r)! \Rightarrow \binom{n}{r} = \frac{n!}{r!(n-r)!}$ □

OBSERVATION

$\forall\, 0 \le r \le n,\ \binom{n}{r} = \binom{n}{n-r}$

Not nice way of doing things 😐

<u>Proof</u>: ① $\binom{n}{n-r} = \frac{n!}{(n-r)!\,(n-(n-r))!} = \frac{n!}{r!(n-r)!} = \binom{n}{r}$ □

bijection

② Choosing a subset of r elements $\leftrightarrow$ Choosing the n-r elements that DON'T go in the subset

LHS options $= \binom{n}{r}$

RHS options $= \binom{n}{n-r}$

$\therefore \binom{n}{r} = \binom{n}{n-r}$ □